Physics 225, Spring 2026: Homework 3 - Solutions
Due 2/12/26

All solutions must clearly show the steps or reasoning you used to arrive at your result. You will lose points for poorly written solutions or incorrect reasoning/explanations; answers given without explanation will not be graded.

Name:
Section:

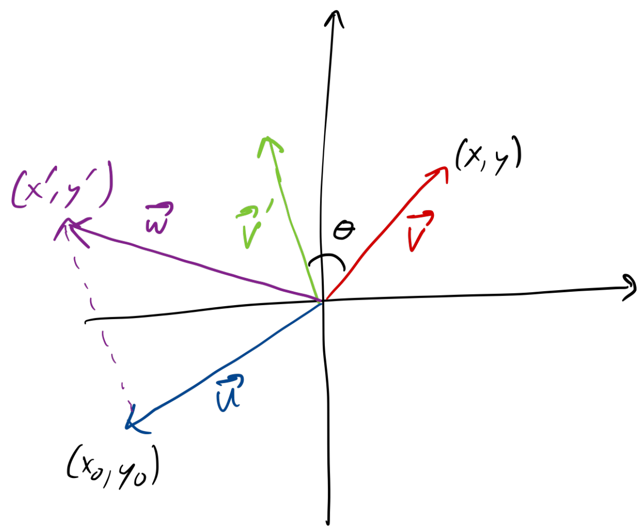
1. **Practice with matrix multiplication and vectors.**

   (a) (10 points) Compute $M\vec{b}$ where $M = \begin{pmatrix} 1 & 0 & 4 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix}$ and $\vec{b} = (-1, 3, 1)$. Show all of your work!
   $M\vec{b} = \begin{pmatrix} 1 \cdot (-1) + 0 \cdot 3 + 4 \cdot 1 \\ 0 \cdot (-1) + 1 \cdot 3 + 2 \cdot 1 \\ 0 \cdot (-1) + 0 \cdot 3 + 1 \cdot 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 5 \\ 1 \end{pmatrix}$.

   (b) (15 points) Problem 1(a) is actually a special case of a very neat trick: we can represent vector *addition* using matrix *multiplication*! Given two 2-dimensional vectors $\vec{a} = (a^1, a^2)$ and $\vec{b} = (b^1, b^2)$, define the 3×3 matrix $M_a = \begin{pmatrix} 1 & 0 & a^1 \\ 0 & 1 & a^2 \\ 0 & 0 & 1 \end{pmatrix}$ and the 3-component vector $\vec{B} = (b^1, b^2, 1)$. Show that the first two components of $M_a\vec{B}$ yields the same result as the vector addition $\vec{a} + \vec{b}$.
   $M_a\vec{B} = \begin{pmatrix} 1 \cdot b^1 + 0 \cdot b^2 + a^1 \cdot 1 \\ 0 \cdot b^1 + 1 \cdot b^2 + a^2 \cdot 1 \\ 0 \cdot b^1 + 0 \cdot b^2 + 1 \cdot 1 \end{pmatrix} = \begin{pmatrix} b^1 + a^1 \\ b^2 + a^2 \\ 1 \end{pmatrix}$. The first two components are $(b^1 + a^1, b^2 + a^2)$ which is identical to the vector addition of $(b^1, b^2) + (a^1, a^2)$.

   (c) (15 points) Compute $A\vec{v}$ where $A = \begin{pmatrix} \cos\theta & -\sin\theta & x^0 \\ \sin\theta & \cos\theta & y^0 \\ 0 & 0 & 1 \end{pmatrix}$ and $\vec{v} = (x, y, 1)$. The result of this matrix multiplication should be a vector $\vec{w} = (x', y', 1)$.
   $A\vec{v} = (x\cos\theta - y\sin\theta + x^0, x\sin\theta + y\cos\theta + y^0, 1)$.

   (d) (20 points) Draw an xy plane and represent $\vec{v}$ as the vector connecting the origin to (x, y), and $\vec{w}$ from part (c) above as the vector connecting the origin to (x', y'). (You may choose some specific values for x, y, x^0, y^0, and θ to do your drawing.) Describe in a sentence or two how $\vec{w}$ is related to $\vec{v}$.



$\vec{w}$ is obtained by rotating $\vec{v}$ counterclockwise by an angle θ to obtain a vector $\vec{v}'$, and then adding $\vec{v}'$ to the vector $\vec{u} = (x^0, y^0)$.

2. **Practice with Einstein summation convention.**

   (a) (10 points) Let $M = \begin{pmatrix} 1 & 3 & 2 \\ 2 & -3 & 1 \\ 0 & 2 & 5 \end{pmatrix}$. Compute M^i_i. Note the repeated upper and lower index means that there is a summation implied! The quantity M^i_i is called the *trace* of the matrix M.

   $M^i_i = M^1_1 + M^2_2 + M^3_3 = 1 + (-3) + 5 = 3.$

   (b) (15 points) Compute δ^i_i, the trace of the $n \times n$ identity matrix, and write your answer in terms of n.

   $\delta^i_i = \sum_{i=1}^n 1 = n$, since we are summing up n 1's along the diagonal of the identity matrix.

   (c) (15 points) The Kronecker delta can be used to compute traces. Show that for any matrix M^i_j, $M^i_j \delta^j_i = M^i_i$.

   δ^j_i is zero unless $i = j$. Therefore, in the implied sum over j in $M^i_j \delta^j_i$, only the term with $j = i$ contributes. We can therefore replace j by i in the index of the matrix M to get M^i_i. Note that we could have done the same with the sum over i, and obtained M^j_j, which is the same as M^i_i because the dummy index label is irrelevant once it is summed over.